x0xb0x Mainboard Beta Release (3/8/2005)

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CPU

File: p02_cpu.kicad_sch

LEDs

File: p05_leds.kicad_sch

VCA

File: p08_vca.kicad_sch

Rotary encoders

File: p03_encoders.kicad_sch

CV and VCO

File: p06_cv_vco.kicad_sch

Envelope

File: p09_env.kicad_sch

Tact switches

File: p04_switches.kicad_sch

VCF

File: p07_vcf.kicad_sch

Mixer and headphone

File: p10_mixer.kicad_sch

x0xb0x mainboard

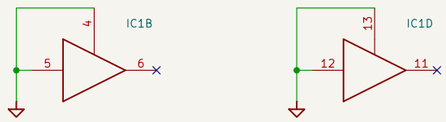
Sheet: /

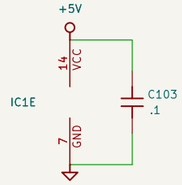
File: mainboard.kicad_sch

Title: x0xb0x mainboard

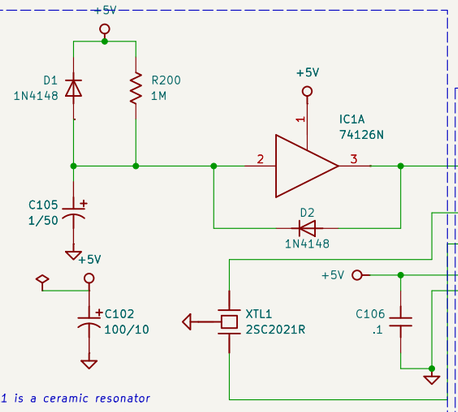
Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 1/10

Unused gates of IC1



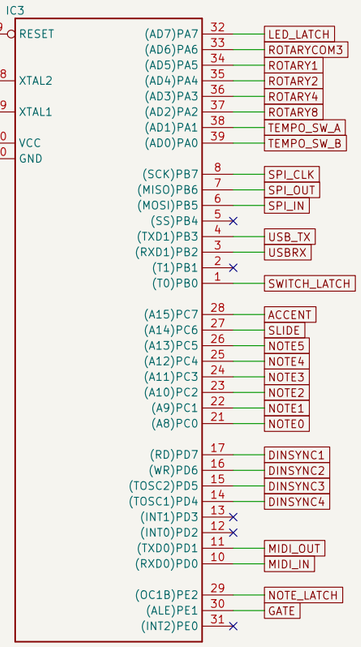


Reset and clock

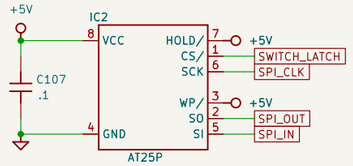


XTL1 is a ceramic resonator

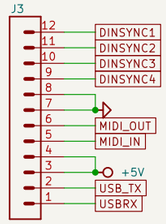
MCU



EEPROM



To the I/O board



Mounting holes



3.3 mm holes: on the board only, not on the original sheet

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Drawn from the original Eagle schematic; nets checked against the board copper.

x0xb0x mainboard

Sheet: /CPU/
File: p02_cpu.kicad_sch

Title: CPU

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 2/10

Rotary Encoders

function select

S4

D6 1N4148
D5 1N4148
D4 1N4148
D3 1N4148

GH3074

R255 10K

Q5 2SC536F

R254 10K

bank select

S3

D10 1N4148
D9 1N4148
D8 1N4148
D7 1N4148

GH3074

ROTARYCOM3

ROTARY8
ROTARY4
ROTARY2
ROTARY1

tempo

J1
TEMPO_SW
S2
J2
+5V
R233 10K
R270 10K
R269 10K
R235 220k
R236 220k
PEC11-S
TEMPO_SW_B
TEMPO_SW_A
C100 .001
C101 .001

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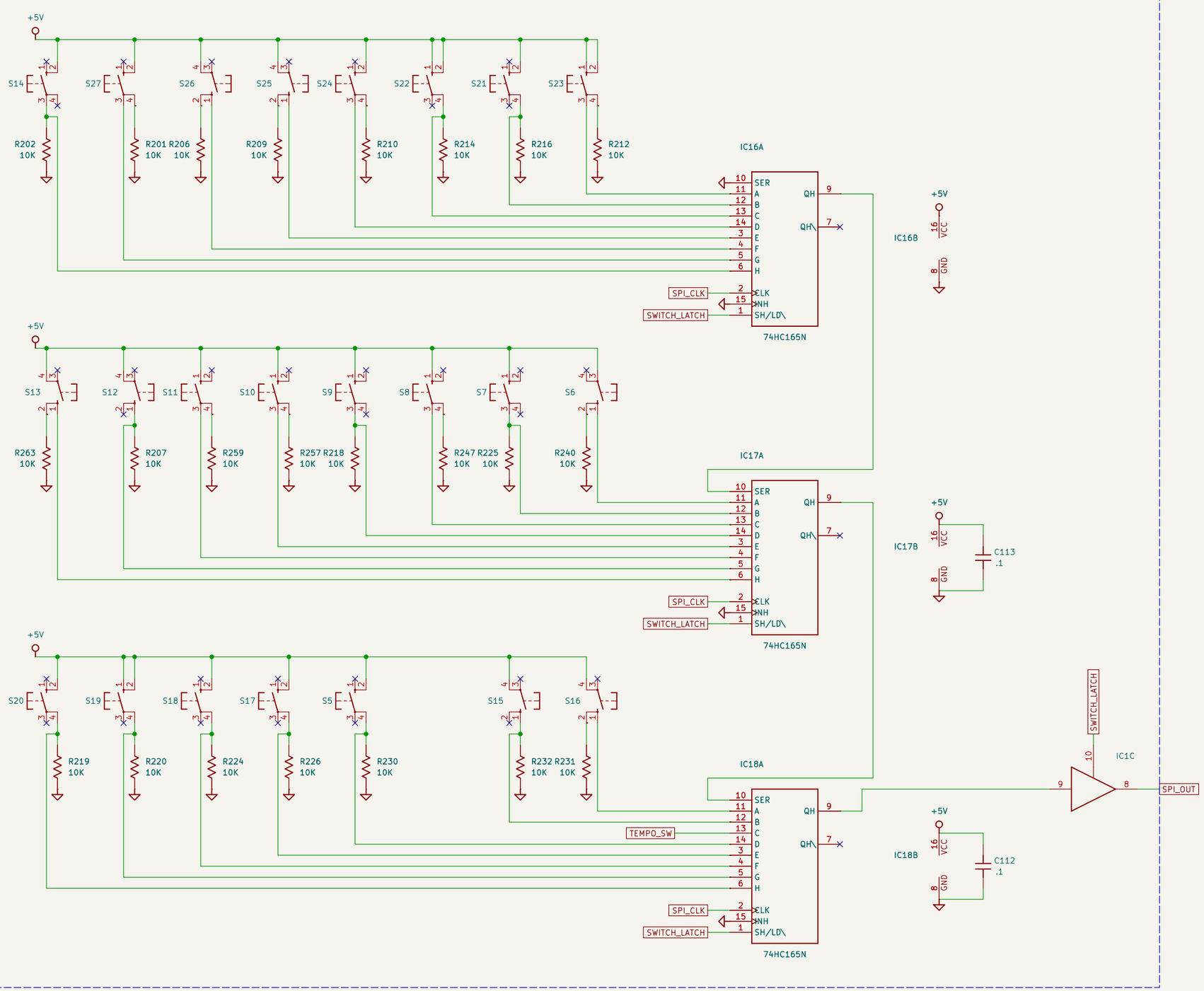
x0xb0x mainboard

Sheet: /Rotary encoders/
File: p03_encoders.kicad_sch

Title: Rotary encoders

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 3/10

Tact Switches



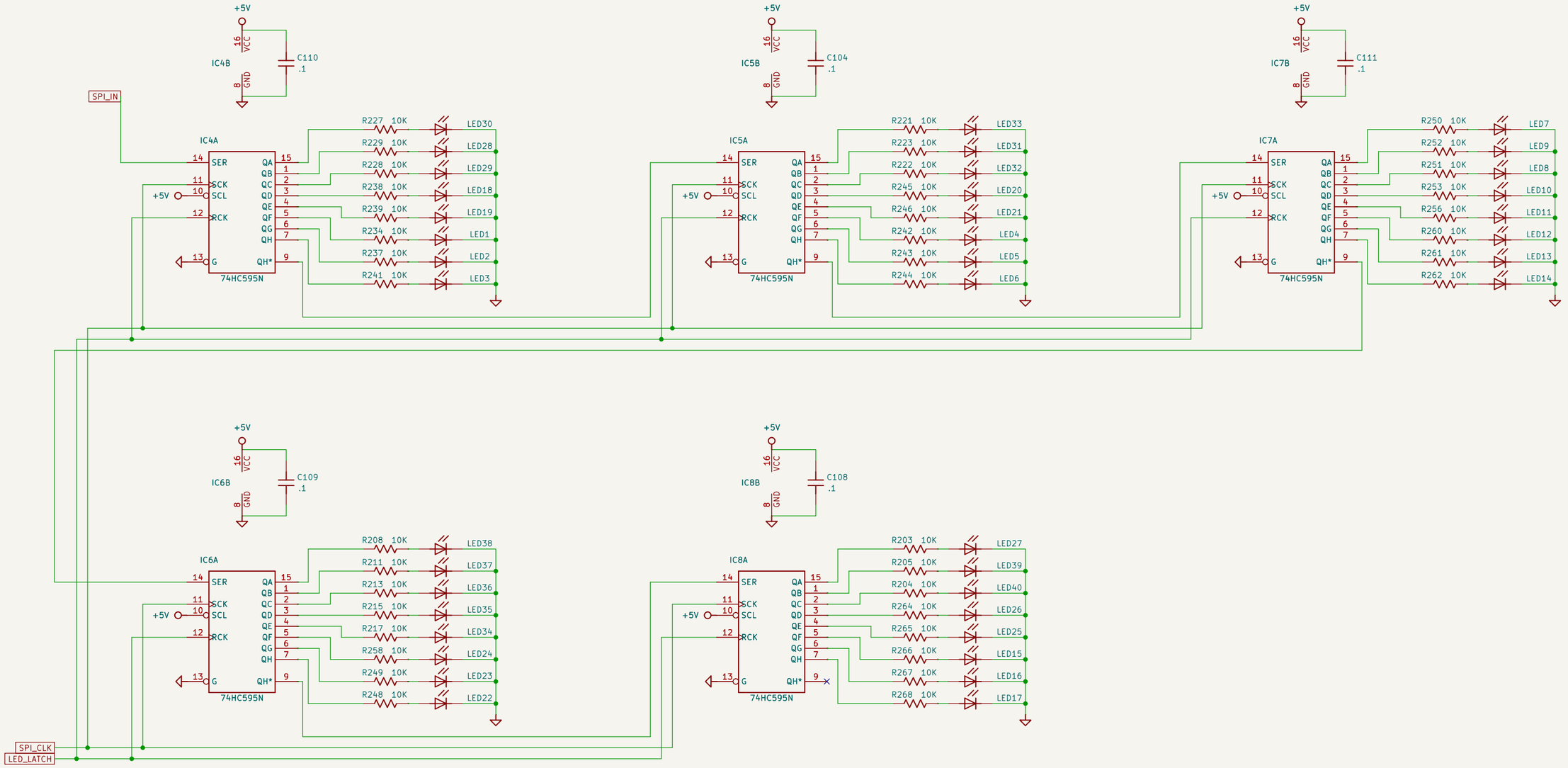
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Drawn from the original Eagle schematic; nets checked against the board copper.
x0xb0x mainboard

Sheet: /Tact switches/
File: p04_switches.kicad_sch

Title: Tact switches

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 4/10

LEDs



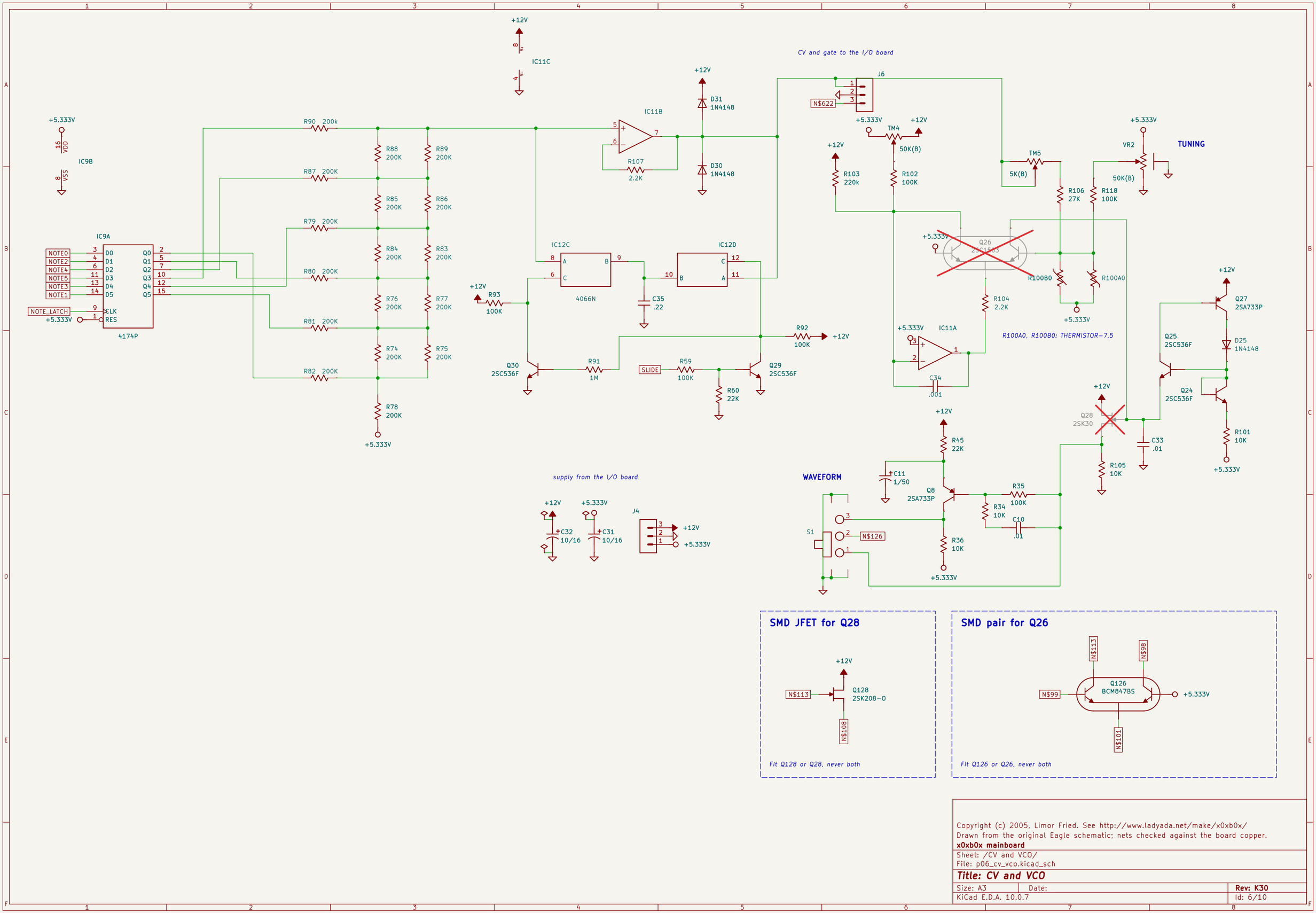
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x0xb0x mainboard

Sheet: /LEDs/
File: p05_leds.kicad_sch

Title: LEDs

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 5/10



SMD JFET for Q28

Fit Q128 or Q28, never both

SMD pair for Q26

Fit Q126 or Q26, never both

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x0xb0x mainboard

Sheet: /CV and VCO/
File: p06_cv_vco.kicad_sch

Title: CV and VCO

Size: A3	Date:	**Rev: K30**
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Diode ladder

R115 10K
Q22 2SC2291
C30 10/16
C26 .033
Q16 2SC536F
Q23 2SC536F
C24 .033
Q14 2SC536F
Q17 2SC536F
C19 .033
Q13 2SC536F
Q15 2SC536F
C18 .018
Q12 2SC1583
from the waveform switch S1
N$126
R62 220K
C17 1/50
R70 2.2K
R98 2.2K
+5.333V
R68 2.2K
C16 10/16
R69 2.2K

Output amplifier and resonance

R95 100
+12V
C28 47/16
R111 22K
R110 22K
Q19 2SC536FUNEVEN
C14 1/50
VR4A
N$27
to C21 and the VCA
N$26
to C20 and the VCA
R96 10K
50K(B)
C25 .1
C27 .1
Q21 2SC1583
R113 100K
R114 100K
+5.333V
R116 10K
RESONANCE
R112 10K
C29 1/50
Q20 2SC536F
R109 10K
R108 2.2K
Q18 2SC536F
R67 2.2K
C15 1/50
R99 100K
R97 10K
C23 1/50
R66 100K
+5.333V
R94 10K
C22 1/50

SMD pairs

Q122 BCM847BS
N$122
N$122
N$143
N$115
N$120

Q112 BCM847BS
N$118
N$121
N$124
N$60
N$127

Q121 BC847CDW1T1G
N$140
N$148
N$138
N$141
N$139

Fit each chip or the SIP part it replaces, never both

Exponential converter

Q11 2SC536F
R73 100K
TM3 500K(B)
+12V
VR3 50K(A)
from MEG
VR5 50K(A)
N$53
CUTOFF FREQ.
Q10 2SA733P
R63 220K
R71 2.2K
R72 100K
ENV. MOD
R61 10K
R47 10K
+5.333V
R65 10K
Q9 2SA733P
R64 10K

Accent sweep

VR4B 50K(B)
C13 1/50
R46 47K
D24 1N4148
N$35
accent line, from R119 and VR7

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x0xb0x mainboard

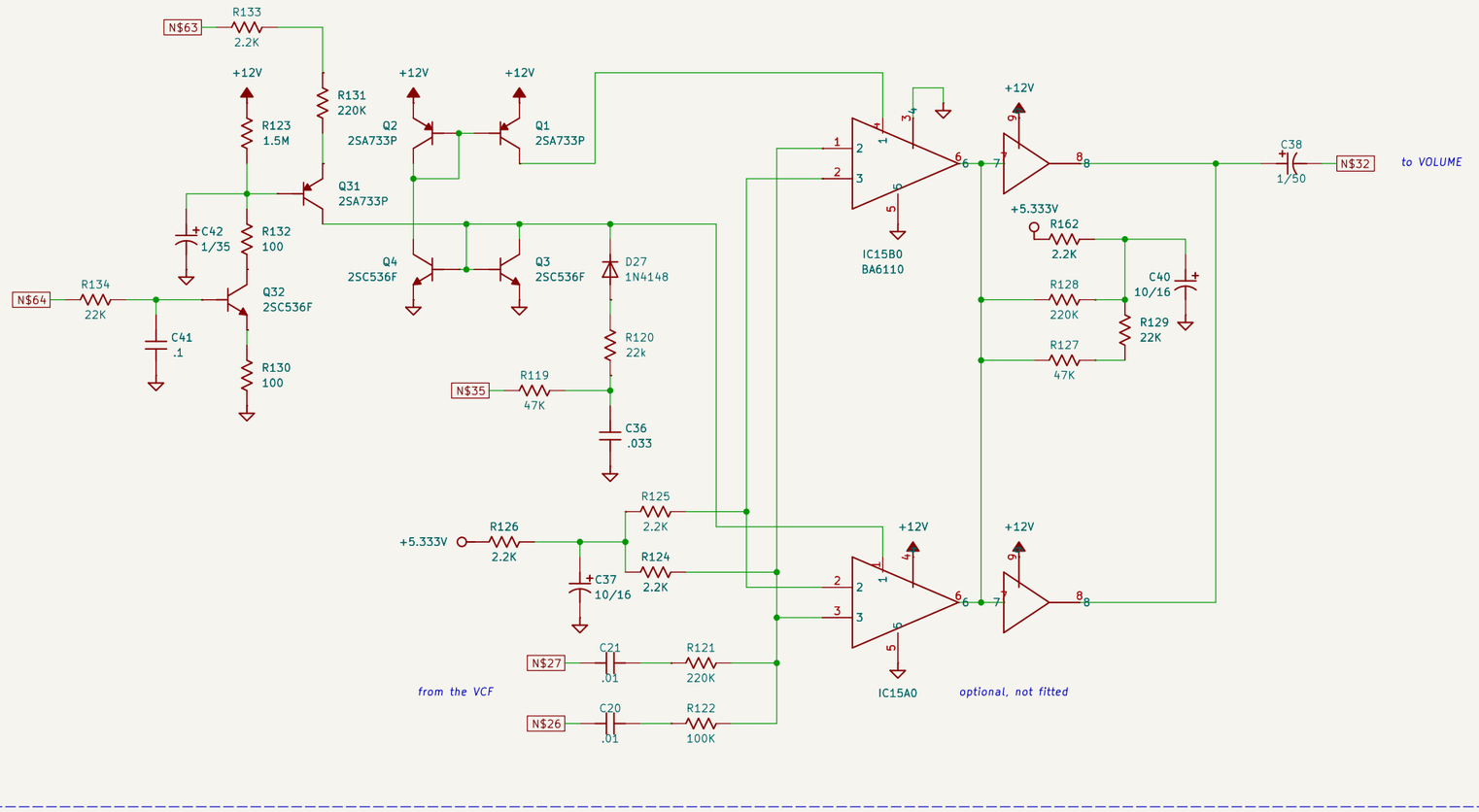
Sheet: /VCF/
File: p07_vcf.kicad_sch

Title: VCF

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 7/10

VCA

the envelope, turned into a current



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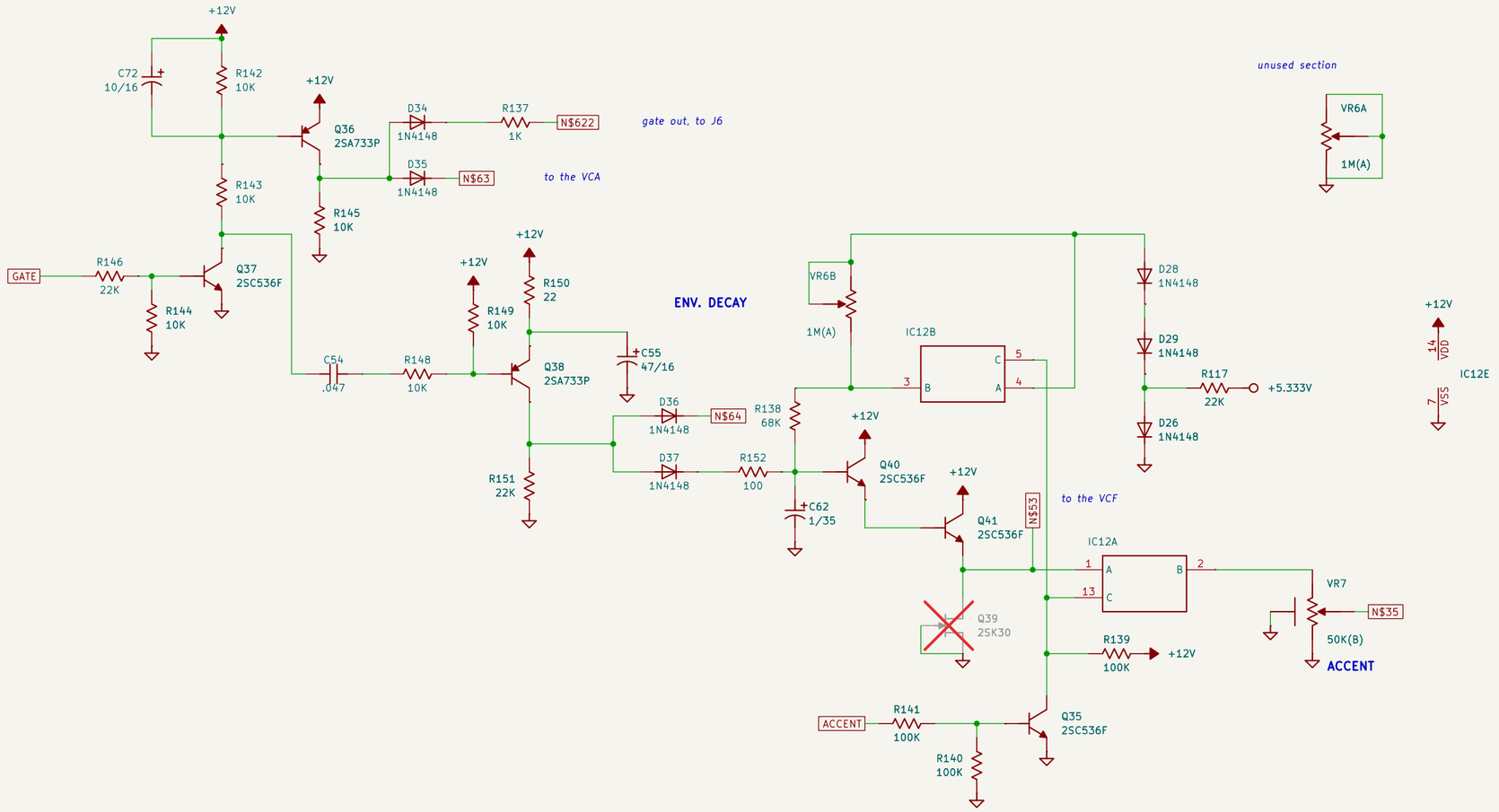
x0xb0x mainboard

Sheet: /VCA/
File: p08_vca.kicad_sch

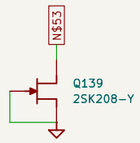
Title: VCA

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 8/10

ENV



SMD JFET for Q39



Fit Q139 or Q39, never both

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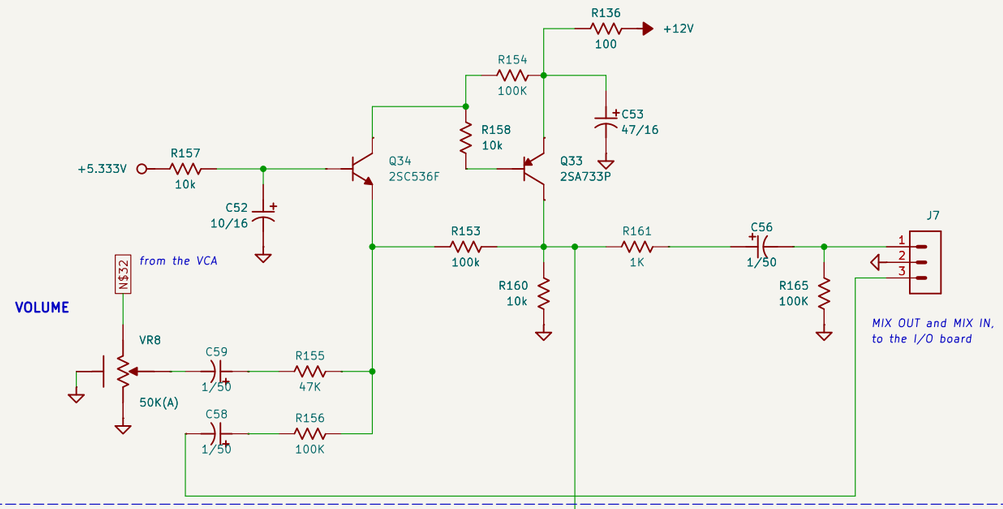
x0xb0x mainboard

Sheet: /Envelope/
File: p09_env.kicad_sch

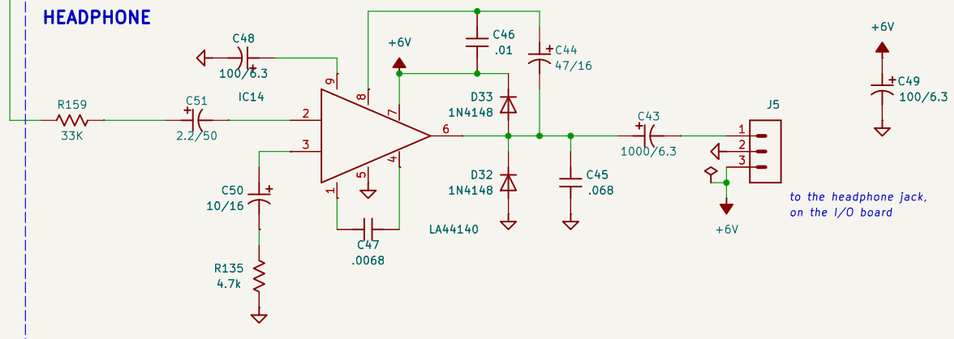
Title: Envelope

Size: A3	Date:	**Rev: K30**
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MIXER



HEADPHONE



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x0xb0x mainboard

Sheet: /Mixer and headphone/
File: p10_mixer.kicad_sch

Title: Mixer and headphone

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 10/10